

Supplementary Methods and Complete Results

VOLT-KD: Amplitude-Preserving Learning
for Lightweight Multi-Label ECG Classification

S1 Comparator Implementation and Measurement Conditions

MSCA-Mamba was implemented from its public methodological description[1]. Three $12 \rightarrow 32$ convolutional branches use kernel 7, dilation rates 1/2/4, batch normalization, ReLU, and squeeze-and-excitation with reduction ratio 4. The concatenated 96 channels pass through a 1×1 fusion convolution, layer normalization, and a unidirectional Mamba module. Mamba uses state dimension 16, local convolution kernel 4, and expansion factor 2. Global average and maximum pooling feed a $192 \rightarrow 128 \rightarrow 64 \rightarrow 5$ classifier, giving 121,149 parameters. Fusion and classifier widths were chosen to satisfy the parameter constraint. Sequence-block and classifier dropout were set locally to 0.1 and 0.3, respectively.

Local MSCA-Mamba input controls used identical networks and training settings. Millivolt inputs were median-centered, whereas z-score inputs were mean-centered and scaled by their standard deviations. VOLT-KD student noise augmentation used standard deviation 0.01 in the respective input units. ECGFounder test predictions averaged logits from four held-out teachers, each containing approximately 30.69 million parameters. Each student training record received soft labels from its corresponding held-out teacher alone.

Efficiency was measured on NVIDIA GB10, with 500 timed GPU repetitions after warm-up and batch size 256 for throughput. CPU measurements used one thread. Timing covered model forward inference, and storage and memory were reported in MiB. Student deployment measurements excluded the training auxiliary head. Teacher computation belonged to the training-stage cost.

S2 Data Structure and Complete Voltage-Probe Results

PTB-XL diagnostic superclasses can co-occur, so a recording may carry several positive labels (Figure S1). Independent binary heads and the multi-label loss allow VOLT-KD to represent these relationships. Voltage probes achieved higher ROC-AUC with millivolt representations on both development data and the test fold (Table S1). Amplitude-related information was therefore present beyond a single test partition.

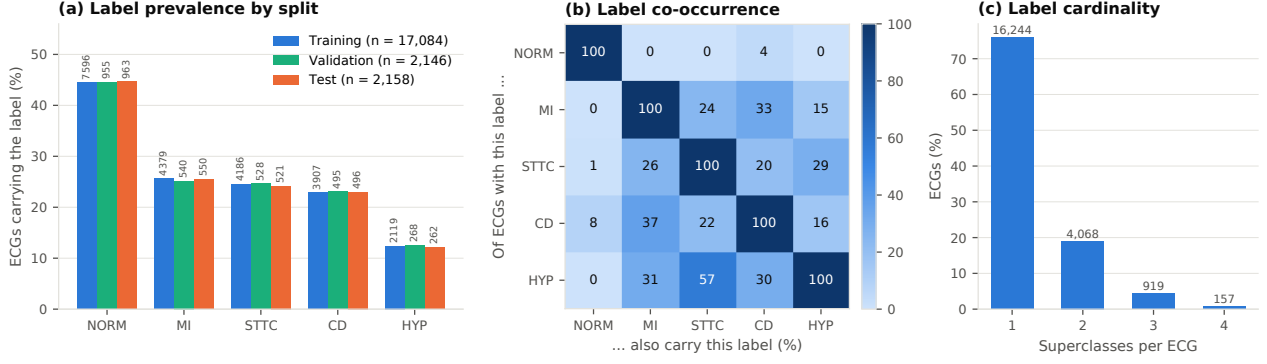


Figure S1: Diagnostic superclass distributions, co-occurrence, and label counts. Splits are patient-disjoint, and labels can co-occur.

Table S1: Training-free voltage-probe ROC-AUC on development data and the test fold. Development data comprise the training and validation folds.

Data	Target	Metric	Positives	mV	z-score
Test fold	HYP	Sokolow-Lyon	262	0.786	0.527
Test fold	HYP	Cornell	262	0.702	0.582
Test fold	LVH	Sokolow-Lyon	214	0.876	0.591
Test fold	LVH	Cornell	214	0.745	0.614
Development	HYP	Sokolow-Lyon	2387	0.809	0.500
Development	HYP	Cornell	2387	0.705	0.543
Development	LVH	Sokolow-Lyon	1918	0.894	0.573
Development	LVH	Cornell	1918	0.723	0.576

S3 Training and Classwise Prediction Behavior

Figure S2 shows five-seed validation trajectories for VOLT-KD and local MSCA-Mamba; all checkpoints were selected using validation data. Classwise ROC, precision-recall curves, and confusion matrices at validation-selected thresholds appear in Figures S3, S4, and S5, respectively. Table S2 reports six classwise metrics. These results characterize both ranking and threshold decisions, retaining class differences that a single macro metric would obscure.

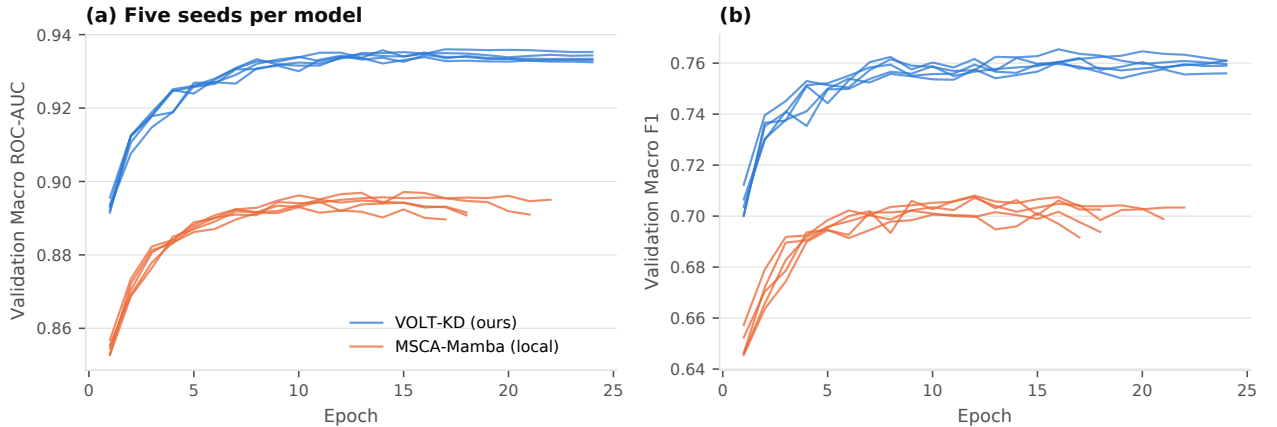


Figure S2: Validation macro ROC-AUC and macro F1 trajectories for complete VOLT-KD and local MSCA-Mamba, each trained with five random seeds.

Table S2: Classwise results of complete VOLT-KD as five-seed means and standard deviations. Bold indicates the highest mean among the five classes in each column.

Class	ROC-AUC	AP	F1	Precision	Recall	Specificity
NORM	0.951 $\pm$ 0.001	0.932 $\pm$ 0.002	0.864 $\pm$ 0.004	0.801 $\pm$ 0.016	0.938 $\pm$ 0.012	0.812 $\pm$ 0.021
MI	0.937 $\pm$ 0.002	0.860 $\pm$ 0.005	0.768 $\pm$ 0.004	0.737 $\pm$ 0.017	0.801 $\pm$ 0.023	0.902 $\pm$ 0.012
STTC	0.942 $\pm$ 0.002	0.837 $\pm$ 0.006	0.779 $\pm$ 0.004	0.741 $\pm$ 0.028	0.823 $\pm$ 0.030	0.908 $\pm$ 0.016
CD	0.925 $\pm$ 0.003	0.859 $\pm$ 0.004	0.768 $\pm$ 0.008	0.835 $\pm$ 0.017	0.712 $\pm$ 0.023	0.958 $\pm$ 0.006
HYP	0.895 $\pm$ 0.002	0.650 $\pm$ 0.011	0.592 $\pm$ 0.008	0.575 $\pm$ 0.054	0.619 $\pm$ 0.065	0.935 $\pm$ 0.020

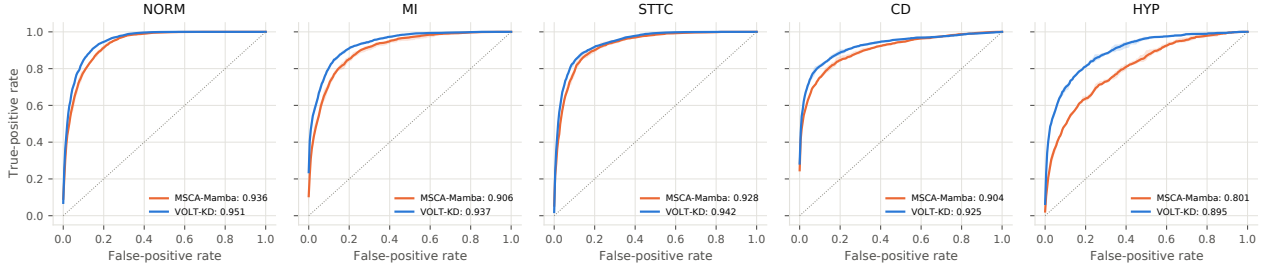


Figure S3: Classwise ROC curves for complete VOLT-KD and local MSCA-Mamba. Solid lines are five-seed means, and shaded regions span the individual-run curves.

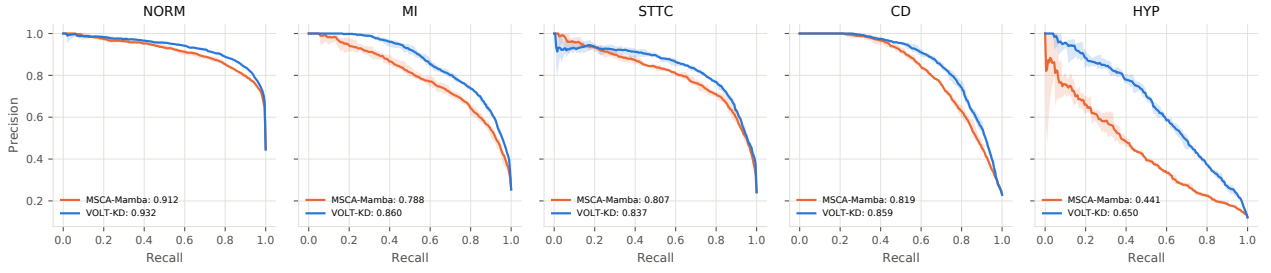


Figure S4: Classwise precision-recall curves for complete VOLT-KD and local MSCA-Mamba.

	NORM (recall 0.893)		MI (recall 0.685)		STTC (recall 0.777)		CD (recall 0.657)		HYP (recall 0.431)			
MSCA-Mamba (local)	True --	992	203	True --	1455	153	True --	1573	89	True --	1755	141
	True +	103	860	True +	173	377	True +	170	326	True +	149	113
	Pred --	Pred +		Pred --	Pred +		Pred --	Pred +		Pred --	Pred +	
	NORM (recall 0.954)		MI (recall 0.796)		STTC (recall 0.787)		CD (recall 0.722)		HYP (recall 0.656)			
VOLT-KD (ours)	True --	938	257	True --	1456	152	True --	1581	81	True --	1760	136
	True +	44	919	True +	112	438	True +	138	358	True +	90	172
	Pred --	Pred +		Pred --	Pred +		Pred --	Pred +		Pred --	Pred +	

Figure S5: Seed-42 binary confusion matrices for local MSCA-Mamba and VOLT-KD. Each class uses its validation-selected threshold.

S4 Additional Literature and Paired-Seed Comparisons

All 45 mean differences between VOLT-KD and the nine published MSCA-Mamba configurations were positive (Figure S6). Comparators span convolutional, recurrent, state-space, and attention architectures, supporting higher aggregate performance within this compact-model set. Intervals were approximated from reported standard deviations and describe seed-run summaries. Published and local case-level predictions were not pooled.

Table S3 separately estimates uncertainty from training randomness and test-sample variation. Paired-seed t intervals use metric differences for common seeds, whereas recording-level bootstrap intervals use 1,000 test-record resamples. Distillation increased macro F1 by 0.0043 beyond subclass supervision, with a seed interval of [0.0022, 0.0064] and a recording interval of [−0.0012, 0.0100]. These results support an F1 gain across repeated training on the fixed test set, while its sample-level increment varies with resampling. Bootstrap tests also report Holm-adjusted results.

Table S3: Paired results for complete VOLT-KD minus local comparator configurations. Except MSCA-Mamba, all comparators are VOLT-KD backbone ablations. AUC, AP, and F1 are macro-averaged.

Comparator	Metric	Difference	95% Seed interval	95% Record interval	p_{Holm}
MSCA-Mamba (local)	AUC	+0.0348	[+0.0328, +0.0368]	[+0.0301, +0.0396]	0.091
MSCA-Mamba (local)	AP	+0.0738	[+0.0695, +0.0782]	[+0.0624, +0.0844]	0.091
MSCA-Mamba (local)	F1	+0.0602	[+0.0568, +0.0635]	[+0.0506, +0.0693]	0.091
Superclass only, z-score	AUC	+0.0192	[+0.0170, +0.0214]	[+0.0158, +0.0229]	0.091
Superclass only, z-score	AP	+0.0480	[+0.0439, +0.0521]	[+0.0376, +0.0572]	0.091
Superclass only, z-score	F1	+0.0353	[+0.0310, +0.0396]	[+0.0279, +0.0431]	0.091
Superclass only, mV	AUC	+0.0020	[+0.0012, +0.0029]	[+0.0002, +0.0037]	1.000
Superclass only, mV	AP	+0.0036	[+0.0002, +0.0071]	[−0.0013, +0.0081]	1.000
Superclass only, mV	F1	+0.0058	[+0.0034, +0.0082]	[+0.0002, +0.0110]	1.000
Superclass + subclass	AUC	+0.0017	[−0.0003, +0.0038]	[+0.0001, +0.0032]	1.000
Superclass + subclass	AP	+0.0034	[−0.0017, +0.0085]	[−0.0009, +0.0072]	1.000
Superclass + subclass	F1	+0.0043	[+0.0022, +0.0064]	[−0.0012, +0.0100]	1.000
Complete supervision z-score	AUC	+0.0119	[+0.0103, +0.0134]	[+0.0089, +0.0151]	0.091
Complete supervision z-score	AP	+0.0325	[+0.0279, +0.0372]	[+0.0230, +0.0415]	0.091
Complete supervision z-score	F1	+0.0259	[+0.0232, +0.0286]	[+0.0186, +0.0331]	0.091
No subclass targets	AUC	−0.0001	[−0.0021, +0.0019]	[−0.0011, +0.0008]	1.000
No subclass targets	AP	−0.0000	[−0.0027, +0.0027]	[−0.0024, +0.0025]	1.000
No subclass targets	F1	+0.0024	[−0.0013, +0.0061]	[−0.0018, +0.0061]	1.000
In-sample teacher	AUC	−0.0006	[−0.0012, −0.0001]	[−0.0019, +0.0004]	1.000
In-sample teacher	AP	−0.0008	[−0.0032, +0.0017]	[−0.0039, +0.0018]	1.000
In-sample teacher	F1	−0.0018	[−0.0079, +0.0043]	[−0.0060, +0.0018]	1.000
noisy-OR head	AUC	−0.0008	[−0.0016, +0.0000]	[−0.0021, +0.0005]	1.000
noisy-OR head	AP	−0.0019	[−0.0137, +0.0099]	[−0.0051, +0.0015]	1.000
noisy-OR head	F1	+0.0008	[−0.0039, +0.0055]	[−0.0040, +0.0063]	1.000
No GRU	AUC	−0.0001	[−0.0007, +0.0005]	[−0.0014, +0.0013]	1.000
No GRU	AP	−0.0019	[−0.0045, +0.0006]	[−0.0051, +0.0011]	1.000
No GRU	F1	+0.0004	[−0.0053, +0.0061]	[−0.0055, +0.0056]	1.000
Half-width student	AUC	+0.0004	[−0.0039, +0.0048]	[−0.0008, +0.0017]	1.000
Half-width student	AP	−0.0016	[−0.0188, +0.0157]	[−0.0052, +0.0016]	1.000
Half-width student	F1	−0.0039	[−0.0137, +0.0059]	[−0.0092, +0.0011]	1.000

The noisy-OR, no-GRU, and half-width VOLT-KD variants use three common seeds; all other configurations use five common seeds. Bootstrap resampling operates on recordings without clustering repeated recordings from the same patient. Paired analyses are exploratory effect estimates.

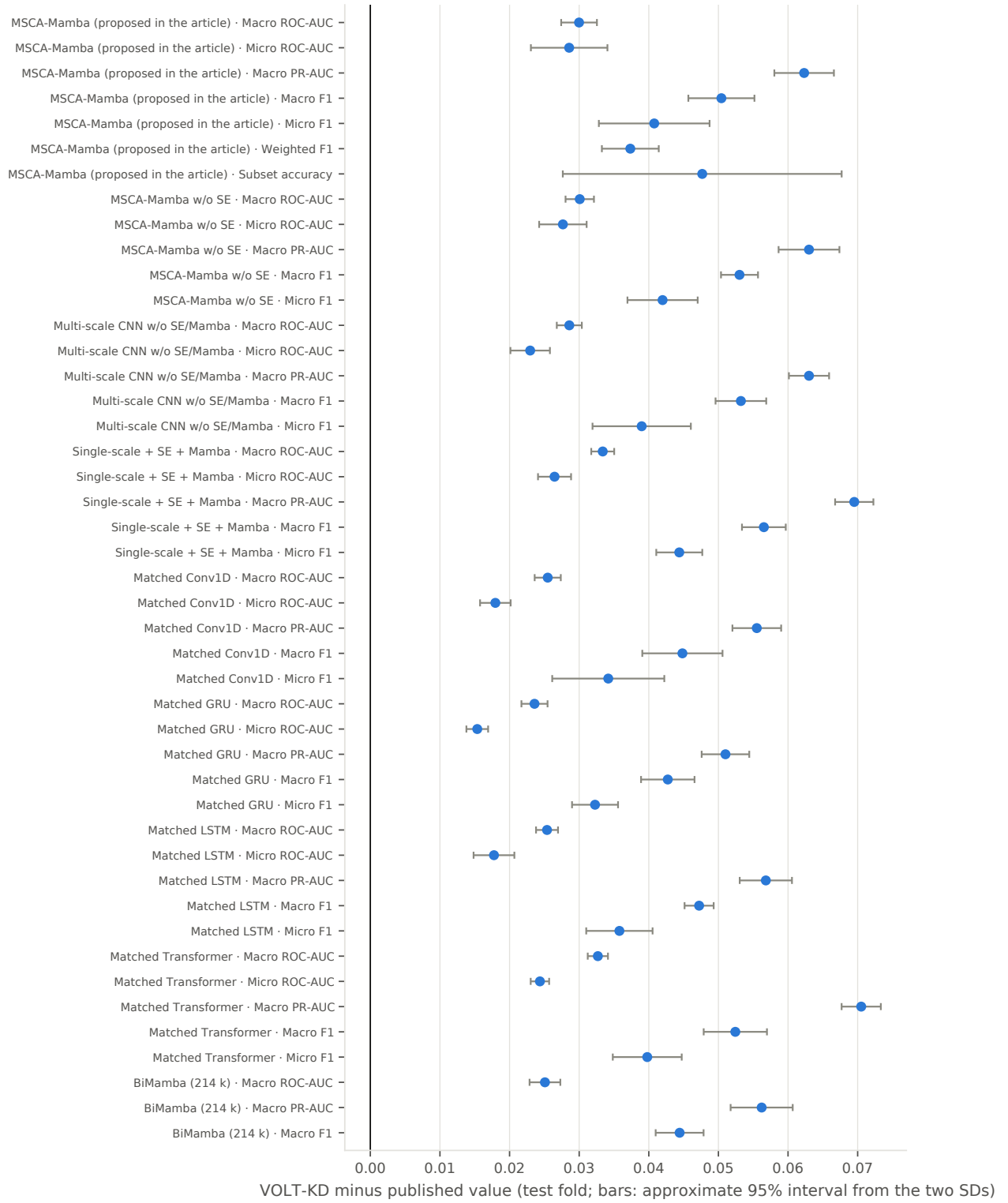


Figure S6: Differences from 45 reported means across nine configurations in the MSCA-Mamba paper[1]. Matched metrics are compared; points are VOLT-KD minus published values, with approximate intervals constructed from reported standard deviations.

S5 Complete Subclass and Subgroup Results

Table S4 reports parent-class detection rates for twenty-one diagnostic subclasses. Improvements for LVH, AMI, IMI, IRBBB, and AVB connect complete VOLT-KD’s multi-label gains to specific diagnostic morphologies. Differences in subclass sample sizes and outcomes are retained. Demographic, device, and recording-condition subgroups use the same test fold (Table S5), with intervals from 300 within-group recording resamples. All test records carry manually validated labels.

Table S4: Parent-class detection rates for complete VOLT-KD and local MSCA-Mamba among diagnostic-subclass records. Subclasses contain at least five recordings; values are five-seed means. Differences use unrounded results.

Parent	Subclass	Records	Local comparator	VOLT-KD	Difference
CD	LAFB/LPFB	179	0.834	0.829	-0.004
CD	IRBBB	112	0.714	0.805	+0.091
CD	AVB	82	0.527	0.644	+0.117
CD	IVCD	79	0.327	0.289	-0.038
CD	CLBBB	54	0.993	0.989	-0.004
CD	CRBBB	54	1.000	1.000	+0.000
CD	ILBBB	8	0.850	0.750	-0.100
CD	WPW	8	0.500	0.450	-0.050
HYP	LVH	214	0.595	0.721	+0.125
HYP	LAO/LAE	42	0.310	0.362	+0.052
HYP	RVH	12	0.133	0.250	+0.117
HYP	RAO/RAE	10	0.120	0.160	+0.040
MI	IMI	327	0.765	0.826	+0.061
MI	AMI	306	0.801	0.854	+0.054
MI	LMI	20	0.380	0.360	-0.020
NORM	NORM	963	0.899	0.938	+0.040
STTC	STTC	222	0.765	0.766	+0.001
STTC	ISC	128	0.925	0.953	+0.028
STTC	ISCA	93	0.832	0.869	+0.037
STTC	NST	77	0.779	0.803	+0.023
STTC	ISCI	40	0.775	0.800	+0.025

Table S5: Test-subgroup comparison of complete VOLT-KD and local MSCA-Mamba. Values are five-seed mean macro ROC-AUC and recording-level bootstrap intervals. Differences are complete VOLT-KD minus local MSCA-Mamba.

Group	Records	MSCA-Mamba (Local comparator)	VOLT-KD (Complete model)	Difference
All records	2158	0.895 [0.887, 0.904]	0.930 [0.922, 0.936]	+0.035
Male	1110	0.906 [0.895, 0.916]	0.940 [0.931, 0.947]	+0.034
Female	1048	0.884 [0.871, 0.896]	0.920 [0.909, 0.930]	+0.036
Age < 40	284	0.894 [0.857, 0.925]	0.930 [0.905, 0.952]	+0.035
Age 40–59	639	0.893 [0.874, 0.911]	0.922 [0.901, 0.940]	+0.030
Age 60–74	706	0.882 [0.867, 0.896]	0.925 [0.913, 0.936]	+0.043
Age ≥ 75	529	0.859 [0.840, 0.880]	0.906 [0.890, 0.919]	+0.047
No noise flag	1645	0.897 [0.888, 0.907]	0.932 [0.925, 0.939]	+0.035
Noise flag	513	0.887 [0.865, 0.902]	0.923 [0.906, 0.937]	+0.036
Single superclass	1650	0.855 [0.840, 0.871]	0.920 [0.907, 0.931]	+0.065
Multiple superclasses	508	0.852 [0.838, 0.865]	0.896 [0.884, 0.908]	+0.044
Device: CS-12	686	0.859 [0.843, 0.877]	0.900 [0.886, 0.915]	+0.040
Device: CS-12 E	437	0.891 [0.863, 0.917]	0.929 [0.908, 0.946]	+0.038
Device: AT-6 C 5.5	382	0.907 [0.893, 0.924]	0.947 [0.934, 0.958]	+0.040
Device: AT-6 6	236	0.905 [0.882, 0.930]	0.944 [0.924, 0.958]	+0.039

S6 Complete Perturbation and Reduced-Lead Results

Table S6 reports macro AUC and standard deviations for eighteen test-perturbation settings. Noise is applied in millivolts before each model’s preprocessing. When training inputs were reduced from twelve leads to six limb leads, MI and HYP showed larger AUC decreases (Table S7 and Figure S7). These changes are consistent with the morphological and voltage information supplied by precordial leads. The cost of lead reduction therefore depends on the observation directions needed by each diagnostic task, not only on the number of leads.

Table S6: Macro ROC-AUC of complete VOLT-KD and local MSCA-Mamba under test perturbations of varying intensity. Values are five-seed means and standard deviations corresponding to Figure 12 in the main manuscript; trained weights remain fixed.

Perturbation	Intensity	MSCA-Mamba (Local comparator)	VOLT-KD (Complete model)
Baseline wander (mV)	0	0.8950 ± 0.0024	0.9298 ± 0.0012
Baseline wander (mV)	0.1	0.8808 ± 0.0033	0.9291 ± 0.0012
Baseline wander (mV)	0.2	0.8507 ± 0.0077	0.9262 ± 0.0014
Baseline wander (mV)	0.5	0.7309 ± 0.0125	0.9058 ± 0.0037
Baseline wander (mV)	1	0.5973 ± 0.0104	0.8032 ± 0.0080
Baseline wander (mV)	2	0.5230 ± 0.0099	0.6070 ± 0.0081
Gaussian noise (mV)	0	0.8950 ± 0.0024	0.9298 ± 0.0012
Gaussian noise (mV)	0.01	0.8949 ± 0.0023	0.9296 ± 0.0012
Gaussian noise (mV)	0.02	0.8937 ± 0.0021	0.9296 ± 0.0011
Gaussian noise (mV)	0.05	0.8841 ± 0.0018	0.9291 ± 0.0009
Gaussian noise (mV)	0.1	0.8499 ± 0.0058	0.9236 ± 0.0011
Gaussian noise (mV)	0.2	0.7720 ± 0.0125	0.8976 ± 0.0041
Lead zeroing	0	0.8950 ± 0.0024	0.9298 ± 0.0012
Lead zeroing	1	0.8853 ± 0.0027	0.9200 ± 0.0015
Lead zeroing	2	0.8714 ± 0.0038	0.9105 ± 0.0018
Lead zeroing	4	0.8315 ± 0.0045	0.8823 ± 0.0010
Lead zeroing	6	0.7678 ± 0.0091	0.8315 ± 0.0027
Lead zeroing	8	0.6763 ± 0.0147	0.7491 ± 0.0084

Table S7: Classwise ROC-AUC for VOLT-KD lead configurations. The complete twelve-lead model uses five seeds; separately trained reduced-lead variants use three seeds.

Input	NORM	MI	STTC	CD	HYP
Twelve leads	0.951 ± 0.001	0.937 ± 0.002	0.942 ± 0.002	0.925 ± 0.003	0.895 ± 0.002
Six limb leads	0.929 ± 0.000	0.888 ± 0.003	0.922 ± 0.001	0.896 ± 0.004	0.848 ± 0.000
Lead II only	0.904 ± 0.002	0.827 ± 0.001	0.885 ± 0.002	0.877 ± 0.003	0.790 ± 0.002

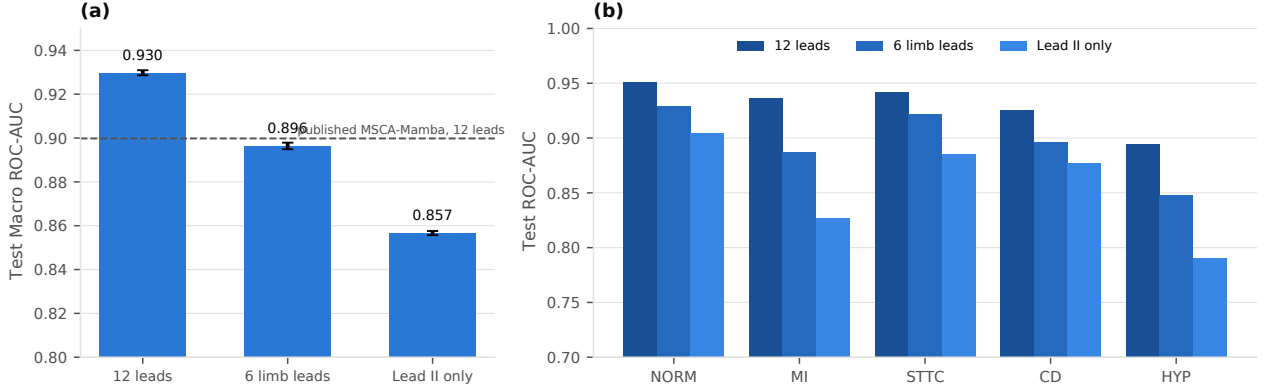


Figure S7: Aggregate and classwise performance of complete twelve-lead VOLT-KD and its six-limb-lead and lead-II student variants. All configurations use twelve-lead ECGFounder soft labels. Aggregate metrics correspond to Table 10 in the main manuscript.

S7 Ensembles, Parameter Configurations, and Local Attribution

Table S8 compares single-model averages with five-seed probability ensembles. The VOLT-KD ensemble achieved macro AUC 0.9326 and AP 0.8338, above single-model averages of 0.9298 and 0.8274. Macro F1 was 0.7541 versus 0.7542, and subset accuracy was 0.6006 versus 0.6168. Ensembling mainly improved ranking, while threshold-based metrics changed in different directions. The primary configuration remains single-model deployment; ensemble results provide a reference under additional computational budget.

Table S8: Single models and five-seed probability ensembles. Single-model rows report means and standard deviations of five independent runs; ensemble rows report one joint evaluation. Bold marks the higher result separately within single-model and ensemble groups.

Configuration	Macro AUC	Macro AP	Macro F1	Subset accuracy
MSCA-Mamba, Single-model mean	0.8950 ± 0.0024	0.7536 ± 0.0033	0.6941 ± 0.0021	0.5618 ± 0.0219
MSCA-Mamba, Five-seed ensemble	0.9031	0.7713	0.7044	0.5820
VOLT-KD, Single-model mean	0.9298 ± 0.0012	0.8274 ± 0.0020	0.7542 ± 0.0009	0.6168 ± 0.0131
VOLT-KD, Five-seed ensemble	0.9326	0.8338	0.7541	0.6006

Parameter count versus AUC reflects the combined resource trade-off of input design and structural simplification (Figure S8). Training counts include the relevant auxiliary branches; deployment counts appear in Table 12 in the main manuscript. Figure S9 shows five correctly classified, high-confidence examples from complete VOLT-KD with seed 42, one per class. The sign of gradient-times-input indicates the local signal’s first-order contribution direction to the current class score. These attributions provide illustrative evidence of waveform use, with interpretation restricted to the displayed records.

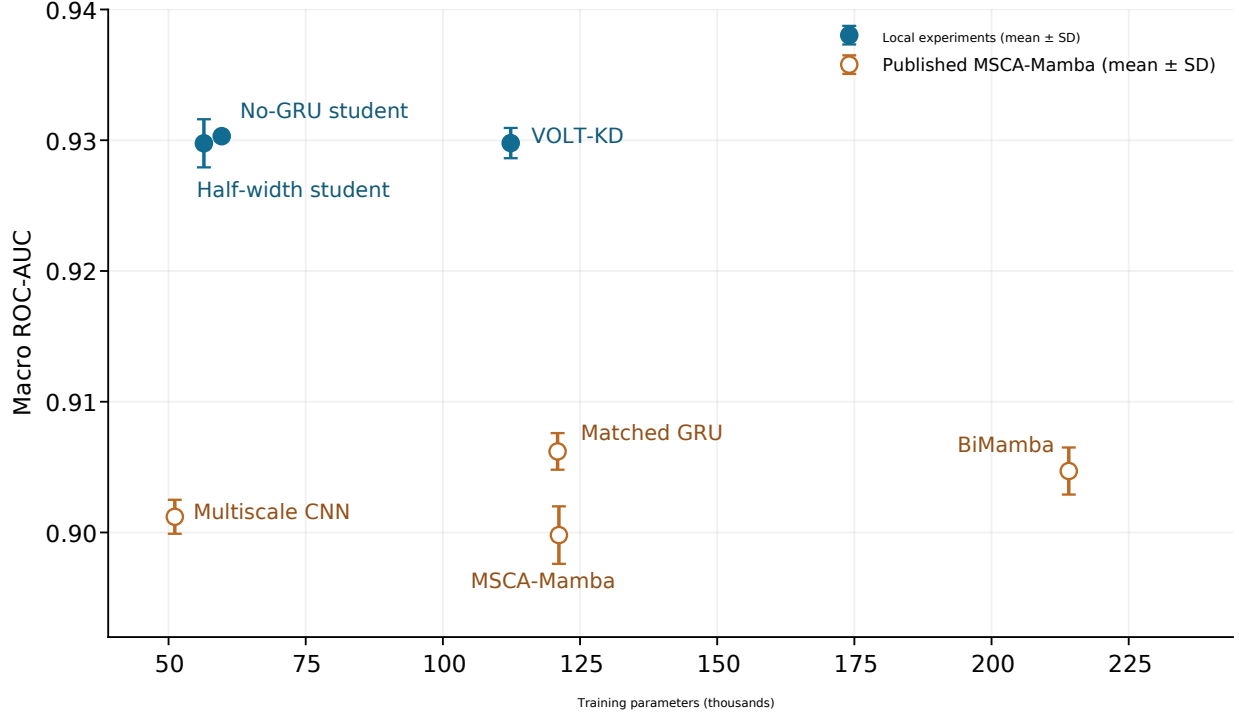


Figure S8: Training parameter counts and macro ROC-AUC of representative compact configurations. Filled points denote complete VOLT-KD, half-width, and no-GRU variants using five, three, and three seeds, respectively. Open points are configurations from the MSCA-Mamba paper[1]; remaining points are local experimental configurations.

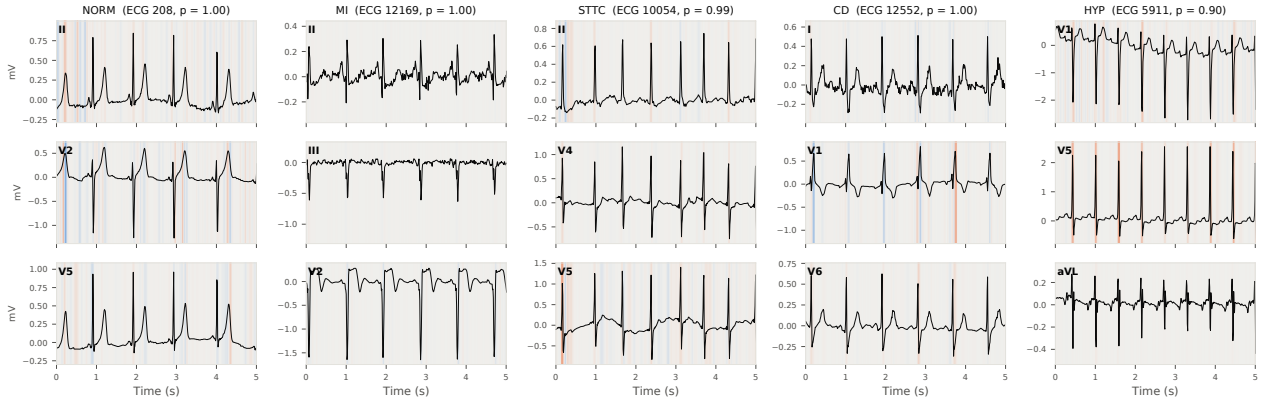


Figure S9: Gradient-times-input attributions for complete VOLT-KD. One correctly classified seed-42 test record per class is shown using three leads. Orange and blue indicate positive and negative contributions to the class score, respectively.

S8 Extended Literature References

The extended comparison covers different information and computational resources (Table S9). Chimera models multivariate time series, DBA-ASFNet combines waveforms and demographics, and CNN-FFT fuses time- and frequency-domain features. Fine-tuned ECGFounder uses 500 Hz inputs, whereas heterogeneous stacking combines several inference models. VOLT-KD uses foundation-model knowledge during training and deploys one 100 Hz student with approximately one hundred thousand parameters. Differences in input information, training knowledge, and inference paths place these methods at distinct accuracy–resource trade-offs. Literature values retain the original authors’ evaluation protocols.

Table S9: Extended literature comparison. All external values are author-reported. Parameters are in millions (M), and bold denotes the best reported value for each metric in this table.

Method and source	Year	Input	Parameters	Macro ROC-AUC	PR-AUC /AP	Macro F1
Chimera[2]	2024	100 Hz	–	0.930	–	–
DBA-ASFNet[3]	2025	100 Hz + age/sex	$\sim$ 0.030	0.9166	–	–
CNN-FFT[4]	2026	500 Hz	0.0869	0.9246	0.813	0.770
Fine-tuned ECGFounder[5]	2026	500 Hz	–	0.930	0.823	0.749
Heterogeneous stacking[5]	2026	Ensemble	–	0.936	0.836	0.763
VOLT-KD (ours)	–	100 Hz	0.1072	0.9298 ± 0.0012	0.8274 ± 0.0020	0.7542 ± 0.0009

Chimera uses the Superdiag result in Table 2 of the NeurIPS 2024 version. DBA-ASFNet uses Super-Diag. in its Table 3 and additionally receives age and sex. CNN-FFT uses its 500 Hz five-superclass results. Both Mert et al. entries come from Table I of the 2026 preprint, using official PTB-XL 1.0.3 folds and seed 42; this study reports five-seed means and standard deviations.

S9 Statistical and Literature-Comparison Conventions

Paired-seed intervals in the main text are computed as $\bar{d} \pm t_{0.975, n-1} s_d / \sqrt{n}$. HYP’s arithmetic share of the macro-AUC gain is defined as $\Delta\text{AUC}_{\text{HYP}} / (5\Delta\text{AUC}_{\text{macro}})$, identifying its contribution to the equally weighted aggregate. All local checkpoints and thresholds are selected using validation data.

Literature comparisons retain the reported precision; dataset versions, inputs, and repeated-run settings appear in Table 3 in the main manuscript and Table S9. MS-LTCAF retains the source’s AUC/F1 terminology, and CNN-FFT uses a 0.5 threshold.

References

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